

Exercise: While Loop

1. Countdown from a Given Number

Write a Java program that uses a while loop to display a countdown from a positive integer to 1.

```
Test Data
Input starting number: 5
Expected Output:
5
4
3
2
1
```

2. Sum of Digits of an Integer

Write a Java program that uses a while loop to find the sum of the digits of a positive integer.

```
Test Data
Input number: 5834
Expected Output:
The sum of the digits is: 20
```

3. Reverse an Integer

Write a Java program that uses a while loop to reverse the digits of an integer.

```
Test Data
Input number: 12345
Expected Output:
The reversed number is: 54321
```

4. Check a Palindrome Number

Write a Java program that uses a while loop to check whether a positive integer is a palindrome.

```
Test Data
Input number: 1221
Expected Output:
1221 is a palindrome number
```

5. Display a Fibonacci Sequence

Write a Java program that uses a while loop to display the first n terms of the Fibonacci sequence.

```
Test Data
Input number of terms: 8
Expected Output:
The Fibonacci sequence is:
0 1 1 2 3 5 8 13
```